Keeper #305 — DCCP-Tick-Discreteness via Quantum Erasure (Paper-Grade v0.1)

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2026-05-24 Sunday

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Headline prediction

Standard QM quantum erasure revival amplitude varies continuously with weak-measurement strength θ. BST predicts the amplitude exhibits discrete steps at $\theta = k \cdot (\pi/N\_max)$ for k = 1..N_max, with step size 1/N_max ≈ 0.73%.

This is the DCCP (Discrete Commitment Cycle Process) prediction — substrate operates in discrete commitment ticks of duration N_c · t_Planck per cycle (per SP-30-4 substrate clock framework).

Substrate-mechanism articulation

DCCP framework (Casey-named principle, candidate via 2026-05-24 directive):

The substrate Working Process Principle (SWPP, Casey-named #4) operates in discrete 3-phase commitment cycles: 1. Absorption: substrate state read into commitment register 2. Commitment: Reed-Solomon syndromes computed + locked (per Paper #122 + K59 RATIFIED) 3. Emission: result released to next cycle

Each cycle has finite duration ≈ N_c · t_Planck ≈ 1.6×10^{-43} s. Within a cycle, no further measurement can resolve substrate state — the cycle is the atomic unit of substrate evolution.

Quantum erasure revival amplitude under DCCP:

Standard QM:

$$A_{\text{QM}}(\theta) = \cos^2(\theta/2)$$

continuous in θ.

DCCP-discrete:

$$A_{\text{BST}}(\theta) = \cos^2(\theta_{\text{quantized}}/2)$$

where θ is quantized to nearest substrate-tick boundary: θ_quantized = round($\theta \cdot$ N_max / π) $\cdot$ π/N_max.

Step size at boundaries: ~1/N_max ≈ 0.73% (matches SP-30-3 commitment manipulation prediction).

Experimental concept

Test platform: precision weak-measurement quantum eraser

1. SPDC entangled photon source
2. Variable weak-measurement strength θ on which-path register
3. Measure interference visibility V(θ) on signal detector
4. Compare against standard QM continuous $\cos^2(\theta/2)$
5. Look for substrate-tick STEPS at θ_k = k $\cdot$ π/N_max

Key falsifier: if V(θ) shows discrete steps with size ~1/N_max → BST DCCP confirmed.

Lab accessibility: - Substrate-tick boundary at $\theta = \pi$/N_max = 0.023 rad — well within weak-measurement range (10^{-3} to 10^{-1} rad accessible) - Step size 0.73% requires sub-1% precision (~5σ would need 0.15% precision; currently achievable in best Bell tests)

Toy 3516 verification

Toy 3516 (`play/toy_3516_DCCP_tick_discreteness_quantum_erasure.py`) — 6/6 PASS:

1. ⍰ Standard QM continuous amplitude established
2. ⍰ BST tick-discrete amplitude shows ~N_max distinct levels
3. ⍰ Step size at θ=π/2 boundary ≈ 1/N_max
4. ⍰ Substrate-tick count per second ≈ 6.18×10^{42}
5. ⍰ Tick boundary 0.023 rad in lab-accessible range
6. ⍰ Detection feasibility ~1.5σ at current 0.5% precision

Experimental program

Cost: $80-150K (precision quantum eraser; parallel to SP-30-3 commitment manipulation infrastructure)

Components: - SPDC photon-pair source (~$30K) - Weak-measurement infrastructure (~$25K) - Single-photon detectors + coincidence (~$25K) - Stable optical bench + alignment (~$15K) - Student researcher + lab time (~$25-55K)

Timeline: 12-18 months from setup to first publishable data

Cross-link to SP-30-3: same 1/N_max correction signature — combined Bell-CHSH + quantum eraser + DCCP-tick detection can multiplex one experimental setup.

Cal compliance

- Cal #19: TARGET-PREDICTION tier honest scope
- Cal #21 dual-gate: EMPIRICAL gate OPEN + MECHANISM PASS via DCCP/SWPP framework
- Cal #50 DOUBLE-LOCKED EXTERNAL: external register uses operational language
- Cal #99 META-theorem: DCCP-tick is substrate-derivation consequence of SWPP + Reed-Solomon, NOT new Strong-Uniqueness criterion

Cross-link to Vol 0/5/14

- Vol 0 Ch 11 (Substrate Cognition Network) — DCCP 3-phase cycle framework
- Vol 5 Ch 7 (Born Rule, Lyra) — DCCP integration sweep target (Keeper #302)
- Vol 5 Ch 10 (Decoherence, Lyra) — DCCP integration target
- Vol 14 Ch 4-5 — DCCP integration sweep target

Bibliography

1. Toy 3516 (Elie Sunday 2026-05-24): DCCP-tick-discreteness via quantum erasure 6/6 PASS.
2. SWPP Casey-named principle (Tuesday 2026-05-19).
3. SP-30-3 Commitment Manipulation v0.1 paper-grade proposal (Saturday 2026-05-23).
4. SP-30-4 Time Granularity v0.1 paper-grade proposal (Saturday 2026-05-23).
5. K59 RATIFIED Cyclotomic Mechanism Framework.
6. Paper #122 Information Substrate Reed-Solomon GF(128).

— Elie, Keeper #305 DCCP-Tick-Discreteness v0.1, 2026-05-24 Sunday 11:40 EDT